SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

Code of Conduct:

I BALAJI S / 192126046 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: B

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Understanding	20%	2	2	2	2
Method Selection	20%	2	2	2	2
Solution Process	25%	2	2	2	2
Accuracy of Calculation	20%	2	2	2	1
Interpretation of Results	15%	1	1	1	1

Total Marks 35 **/40**

Signature of the Faculty

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method

Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

SIMATS ENGINEERING

ASSESSMENT TOOL 1 — CSA07: COMPUTER NETWORKS

Course Outcome: CO2 (BL3) **Tool Weightage:** 40%

Code of Conduct: I certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
Student Name / Reg No: _____ **Signature:** _____

Question 1: Fixed & Variable Subnet Allocation

A university network is assigned IP block `192.168.20.0/24`. Divide this network into subnets for three labs: **Lab A** (/26), **Lab B** (/27), and **Lab C** (/28). Determine network addresses, valid host ranges, broadcast addresses, and total usable hosts for each.

General Subnetting Formulas:

Host bits (h) = 32 − Prefix Length | Block Size = 2^h | Usable Hosts = $2^h - 2$

Step-by-Step Subnet Allocations:

- **Lab A (/26):** Subnet Mask = `255.255.255.192`. Host bits h = 6. Block Size = $2^6 = 64$.
  • Network Address: `192.168.20.0` | Broadcast: `192.168.20.63`
  • Valid Host Range: `192.168.20.1` to `192.168.20.62` (Usable: **62**)
- **Lab B (/27):** Subnet Mask = `255.255.255.224`. Host bits h = 5. Block Size = $2^5 = 32$.
  • Network Address: `192.168.20.64` | Broadcast: `192.168.20.95`
  • Valid Host Range: `192.168.20.65` to `192.168.20.94` (Usable: **30**)
- **Lab C (/28):** Subnet Mask = `255.255.255.240`. Host bits h = 4. Block Size = $2^4 = 16$.
  • Network Address: `192.168.20.96` | Broadcast: `192.168.20.111`
  • Valid Host Range: `192.168.20.97` to `192.168.20.110` (Usable: **14**)

Lab	Subnet Mask	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
Lab A	255.255.255.192 (/26)	192.168.20.0	192.168.20.1 – 192.168.20.62	192.168.20.63	**62**
Lab B	255.255.255.224 (/27)	192.168.20.64	192.168.20.65 – 192.168.20.94	192.168.20.95	**30**
Lab C	255.255.255.240 (/28)	192.168.20.96	192.168.20.97 – 192.168.20.110	192.168.20.111	**14**

Question 2: Host Configuration & Membership Analysis

> System 1: IP `192.168.1.130`, Mask `255.255.255.192`. System 2: IP `192.168.1.190`, Mask `255.255.255.192`. Determine subnet range, network address, broadcast address, valid host IP range, and membership for both systems.

Subnet Mask Analysis: `255.255.255.192` = /26. Host bits h = 32 − 26 = 6. Block size = 2^6 = 64. Usable hosts = 62.

- **System 1 (192.168.1.130):** 4th octet binary AND: 130 (10000010) AND 192 (11000000) = 128 (10000000).
  • Network: `192.168.1.128` | Broadcast: `192.168.1.191` | Host Range: `192.168.1.129 – 190`.
  • *Validation:* 130 is within [129, 190]. System 1 is a valid host in this subnet.
- **System 2 (192.168.1.190):** 4th octet binary AND: 190 (10111110) AND 192 (11000000) = 128 (10000000).
  • Network: `192.168.1.128` | Broadcast: `192.168.1.191` | Host Range: `192.168.1.129 – 190`.
  • *Validation:* 190 is within [129, 190]. System 2 is the last valid host in this subnet.

System	Configured IP	Network Address	Broadcast Address	Valid Host Range	In Subnet?
System 1	192.168.1.130	192.168.1.128	192.168.1.191	192.168.1.129 – 190	**Valid Host**
System 2	192.168.1.190	192.168.1.128	192.168.1.191	192.168.1.129 – 190	**Valid Host**

Key Finding: Both systems share the exact same subnet (192.168.1.128/26) and can communicate directly.

Question 3: Organizational Address Allocation

> Allocated IP Block: `172.16.0.0/16`. Department Subnets: **HR** (/25), **Sales** (/26), **IT** (/27), **Admin** (/28). Assign subnets sequentially, determine ranges, usable host counts, and remaining unused space.

Department	Prefix	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
HR	/25	172.16.0.0	172.16.0.1 – 172.16.0.126	172.16.0.127	**126**
Sales	/26	172.16.0.128	172.16.0.129 – 172.16.0.190	172.16.0.191	**62**
IT	/27	172.16.0.192	172.16.0.193 – 172.16.0.222	172.16.0.223	**30**
Admin	/28	172.16.0.224	172.16.0.225 – 172.16.0.238	172.16.0.239	**14**

Unused Address Space: `172.16.0.240` to `172.16.255.255` (Total 65,296 addresses remaining).

Question 4: Subnet Identification & Target Verification

> Base IP: `172.16.10.0/24` with `/27` subnet mask. Calculate subnets created, usable hosts per subnet, locate IP `172.16.10.78`, and verify if `172.16.10.95` belongs to the same subnet.

- **Subnets & Host Capacity:** Borrowed bits s = 27 − 24 = 3. Remaining host bits h = 32 − 27 = 5.
  - Subnets created = 2^3 = **8 subnets**.
  - Usable hosts/subnet = $2^5 - 2 = 32 - 2$ = **30 usable hosts**. Block size = 32.
- **Subnet for 172.16.10.78:** Index = $\lfloor 78 / 32 \rfloor = 2$.
  - Network Address = 2 × 32 = `172.16.10.64`.
  - Broadcast Address = 64 + 32 − 1 = `172.16.10.95`.
  - Valid Host Range = `172.16.10.65` to `172.16.10.94`.
- **Verification of 172.16.10.95:** Falls in range [64, 95]. It is the **Broadcast Address** for this subnet (cannot be assigned to a host).

Parameter	Value / Result
Total Subnets Created	**8 Subnets** (2^3)
Usable Hosts per Subnet	**30 Hosts** ($2^5 - 2$)
Subnet for 172.16.10.78	172.16.10.64 /27
Network / Broadcast Address	Network: 172.16.10.64 | Broadcast: 172.16.10.95
Status of 172.16.10.95	Belongs to same subnet as **Broadcast Address**

Question 5: IPv6 Addressing, Compression & Prefix Math

> Given IPv6 Address: `2001:0db8:0000:0000:0000:ff00:0042:8329` with prefix `/64`. Compress to shortest form, expand back to full notation, find network prefix, and calculate total host capacity.

- **Compression (RFC 5952):** Omit leading zeros in hextets (`0db8` → `db8`, `0042` → `42`). Replace continuous zero sequence (`0:0:0`) with `::`.
  - Compressed Address: `2001:db8::ff00:42:8329`
- **Expansion:** Restore missing 3 zero blocks (8 total − 5 present): `2001:0db8:0000:0000:0000:ff00:0042:8329`
- **Network Prefix (/64):** First 64 bits (4 hextets) → `2001:db8::/64`
- **Total Possible Host Addresses:** Host bits h = 128 − 64 = 64 bits.
  - Capacity = 2^{64} = **18,446,744,073,709,551,616 addresses** ($\approx 1.84 \times 10^{19}$).

IPv6 Parameter	Result Value
Shortest Compressed Form	`2001:db8::ff00:42:8329`
Full Expanded Notation	`2001:0db8:0000:0000:0000:ff00:0042:8329`
Network Prefix (/64)	`2001:db8::/64`
Total Host Capacity	**2^{64} = 18,446,744,073,709,551,616**